Experiment 7

Subnetting & IP Assignment

Lab 7 Objectives:

● Perform subnetting calculations and assign appropriate IPs to networked systems.

Prerequisites: The prerequisites of this lab are-

● Basic understanding of IP addressing and subnetting.
● Familiarity with binary conversion and subnet mask calculations.

Outcomes: By the end of this lab, students will be able to:

● Calculate subnet masks and determine network and host ranges.
● Assign IP addresses to devices based on subnet requirements.
● Verify network connectivity and troubleshoot IP conflicts.

Step 1: Determine Subnet Requirements

Given network: **192.168.10.0/24**
Requirement: Divide this network into **4 subnets** with at least **50 hosts** per subnet.

- Number of subnets required = 4
- Minimum hosts per subnet = 50

Step 2: Calculate the Subnet Mask

Default Information

- The IP address 192.168.10.0 is a **Class C** IP address.
- Default subnet mask for Class C is **255.255.255.0** or **/24**.
- This means 8 bits are available for host addresses (last octet).

Calculate number of bits to borrow for subnetting
To create 4 subnets:

$$\underline{2^n \geq 4 \Longrightarrow n = 2 \text{ bits borrowed}}$$

- Borrow 2 bits from the host portion for subnetting.
- Remaining bits for hosts = 8 - 2 = 6 bits.

Calculate maximum hosts per subnet

$$\underline{2^6 - 2 = 64 - 2 = 62 \text{ usable hosts per subnet}}$$

(2 addresses are reserved for network address and broadcast address)
Since 62 ≥ 50, this subnet mask satisfies the host requirement.

Calculate the new subnet mask

- Borrowing 2 bits means the new subnet mask is:

  $255.255.255.(256 - 2^{8-2}) = 255.255.255.192$

- In CIDR notation, this is **/26**.

Step 3: Find Subnet Addresses and Usable IP Ranges

Each subnet has 64 IP addresses (including network and broadcast addresses).

Subnet	Network Address	Usable Host IP Range	Broadcast Address
1	192.168.10.0	192.168.10.1 – 192.168.10.62	192.168.10.63
2	192.168.10.64	192.168.10.65 – 192.168.10.126	192.168.10.127
3	192.168.10.128	192.168.10.129 – 192.168.10.190	192.168.10.191
4	192.168.10.192	192.168.10.193 – 192.168.10.254	192.168.10.255

Step 4: Assign IP Addresses to Devices

Assign unique IP addresses within the usable range to each device on a subnet. For example:

Device	Subnet	Assigned IP	Subnet Mask
PC1	1	192.168.10.1	255.255.255.192 (/26)
PC2	1	192.168.10.2	255.255.255.192 (/26)
PC3	2	192.168.10.65	255.255.255.192 (/26)
PC4	3	192.168.10.130	255.255.255.192 (/26)

Step 5: Configure and Verify IP Settings

- Configure IP addresses and subnet masks on devices.
- To check IP configuration on Windows, use the command:

<u>ipconfig /all</u>

```
Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Intel(R) Wi-Fi 6E AX211 160MHz
   Physical Address. . . . . . . . . : DC-45-46-A0-10-13
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
   IPv4 Address. . . . . . . . . . . : 192.168.226.42(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Lease Obtained. . . . . . . . . . : 18 May 2025 17:17:47
   Lease Expires . . . . . . . . . . : 18 May 2025 19:48:13
   Default Gateway . . . . . . . . . : 192.168.226.177
   DHCP Server . . . . . . . . . . . : 192.168.226.177
   DNS Servers . . . . . . . . . . . : 192.168.226.177
   NetBIOS over Tcpip. . . . . . . . : Enabled

Ethernet adapter Bluetooth Network Connection:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Bluetooth Device (Personal Area Network)
   Physical Address. . . . . . . . . : DC-45-46-A0-10-17
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
```

Use tracert (Windows)

```
C:\Users\CHETAN>tracert 192.168.226.42

Tracing route to LAPTOP-TIQMNKVG [192.168.226.42]
over a maximum of 30 hops:

  1    <1 ms    <1 ms    <1 ms  LAPTOP-TIQMNKVG [192.168.226.42]

Trace complete.
```

Step 6: Troubleshoot IP Conflicts and Connectivity Issues

If ping fails, check the following:

- IP address and subnet mask are configured correctly.
- No duplicate IP addresses on the network.
- Network cables or virtual links are connected properly.
- Devices are powered on and reachable.